UNCOVERING CUSTOMER BEHAVIOUR PATTERNS: A MULTI-DIMENSIONAL K-MEANS CLUSTERING APPROACH WITH VALIDATION TECHNIQUES

STUDENT ID: 24155168

GitHub: https://github.com/porlasriya/Assignment

Introduction:

Understanding consumer behaviour is important for businesses that want to enhance marketing strategies and customer engagement. In retail settings, customers are known to spend differently as a result of their what they earn, age and much more. Uncovering significant patterns from such data can allow for useful and timely decisions.

Clustering methods is a powerful approach to reveal hidden structures in data without prior labeling. K-Means clustering is one of the simplest yet effective ways of clustering observations into groups. Deciding the right number of clusters and creating meaningful segments are key issues.

In this tutorial, you'll learn how to apply the K-Means clustering algorithm to segment customers based on their purchasing behaviour. The Elbow Method, analysing the Silhouette Score, and visualising in multi-dimensions will help in finding the optimal clusters and their interpretation. Inclusion of a three-dimensional analysis helps the project to achieve a better insight on Customer Segmentation.

Background:

→**Unsupervised Learning and Clustering:** Clustering is a method of unsupervised learning, and identification of patterns is the main goal in this method. Clustering algorithms, unlike supervised models, do not follow any pre-defined external criteria. The main goal is to categorize observations such that any observation in a cluster is much similar to the remaining observations in the cluster than to other clusters.

→**K-Means Clustering:** K-Means clustering divides the data into k different groups or clusters to minimize WCSS The algorithm iteratively assigns each data point to the nearest centroid and updates the centroid positions until convergence is achieved. The optimization goal ensures that the clusters are compact and consistent.

K-Means is simple, yet very only effective due to a feature scaling and a selected number of clusters. Hence, more methods are needed for effective results.

→**Feature Scaling:** In K-Means this distance calculation can cause a feature with a larger range to dominate clustering. Standardisation scales the variables to a common range so they contribute equally. For balanced and meaningful clustering, this step is essential.

→**Multi-Dimensional Clustering:** While clustering is often visualised in two dimensions, real-world data is inherently multi-dimensional. Incorporating additional features allows for a richer representation of relationships within the dataset. This study includes a three-dimensional analysis to capture more complex patterns.

→**Cluster Validation Techniques:** Choosing the right number of clusters is an important step in cluster analysis.

- The Elbow Method analyzes the reduction of WCSS with the increase in clusters. The best point is when adding more clusters yields less and less improvement.
- A metric for how similar an object is to its own cluster to other clusters is referred to as Silhouette score. Better-defined clusters have higher values.

Employing both techniques collectively aids in assuring the cluster configuration.

Methodology

→**Dataset Description:** The dataset consists of mall customer information, including age, annual income, and spending score. These variables capture both demographic and behavioural aspects of customer activity. For the primary analysis, income and spending score were selected due to their direct relevance to purchasing behaviour.

→**Data Preprocessing:** Prior to clustering, the selected features were standardised using feature scaling. This ensured that income and spending score contributed equally to distance calculations, preventing bias in cluster formation.

→**Model Implementation:** The K-Means algorithm was applied to the scaled dataset. The number of clusters was initially unknown and determined using validation techniques. The model was trained with different values of k to evaluate clustering performance.

→**Validation Strategy:** The Elbow Method and Silhouette Score were used to determine the optimal number of clusters. The results from both methods were compared to ensure consistency and reliability.

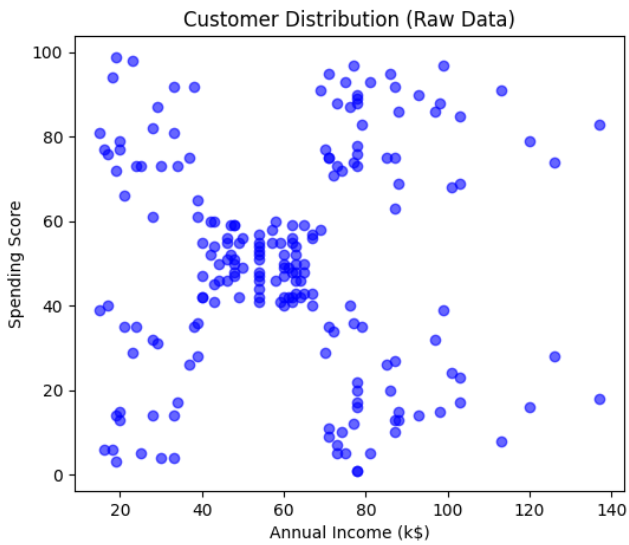
→**Extended Analysis:**

To enhance the depth of the study, the following analyses were conducted:

- Visualisation of customer distribution
- Cluster formation in both scaled and original feature space
- Cluster profiling based on average values
- Distribution of customers across clusters
- Three-dimensional clustering incorporating age

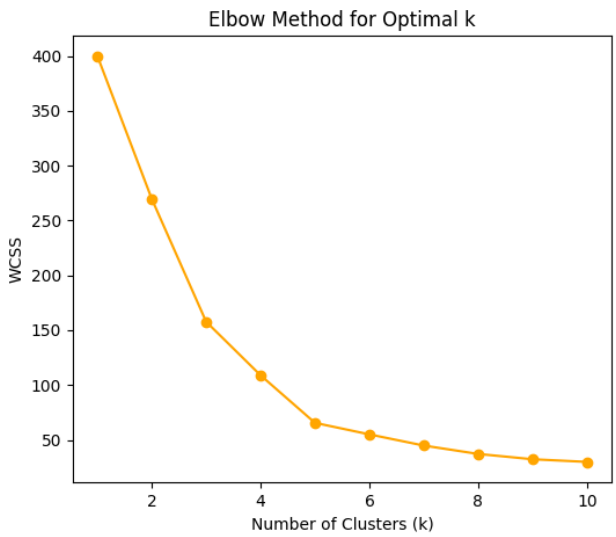
Results and Visual Analysis

Customer Distribution: The scatter plot of income versus spending score reveals a non-uniform distribution of customers. Several dense regions are visible, suggesting the presence of natural groupings. This observation supports the application of clustering techniques for segmentation.

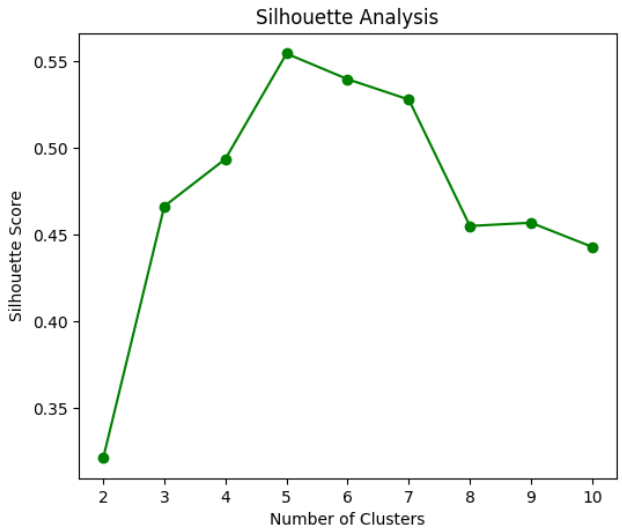


Elbow Method:

The Elbow Method shows a sharp decline in WCSS up to k = 5, after which the rate of improvement slows significantly. This indicates that five clusters capture the essential structure of the dataset without unnecessary complexity.



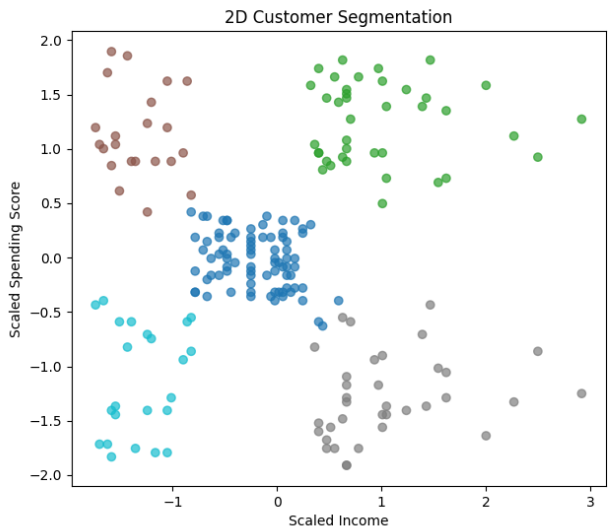
Silhouette Analysis:



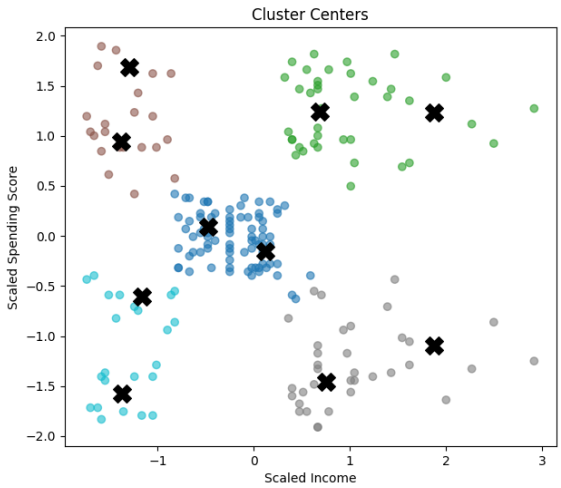
The silhouette score peaks around five clusters, confirming that this configuration provides well-separated and cohesive clusters. The agreement between validation methods strengthens the reliability of the chosen model.

Two-Dimensional Clustering:

The 2D clustering results show clear separation between customer groups. Each cluster represents a segment with distinct income and spending characteristics. The visual clarity of separation indicates that the clustering model is effective.



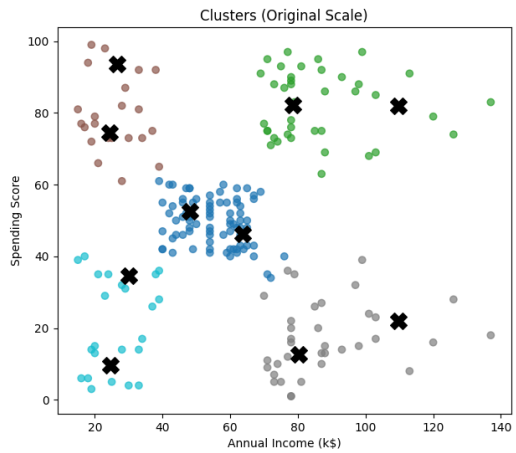
Cluster Centres:



The centroids represent the average characteristics of each cluster. Their positions provide insight into typical customer behaviour within each group. The clear separation of centroids confirms that clusters are well-defined.

Clusters in Original Scale:

Transforming the clusters back to the original scale improves interpretability. This allows the results to be understood in real-world terms, making it easier to derive business insights

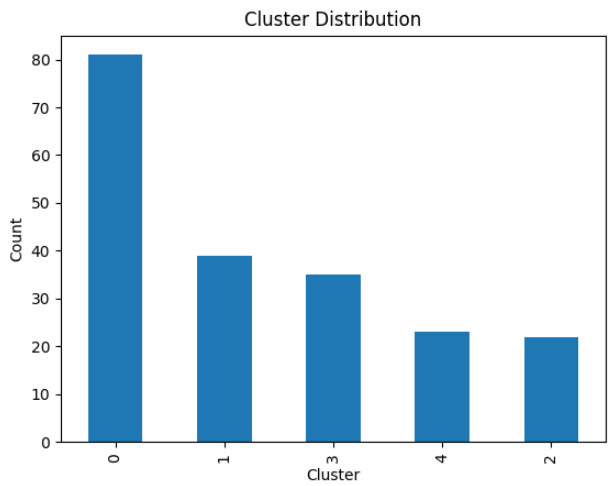


Cluster Profiles:

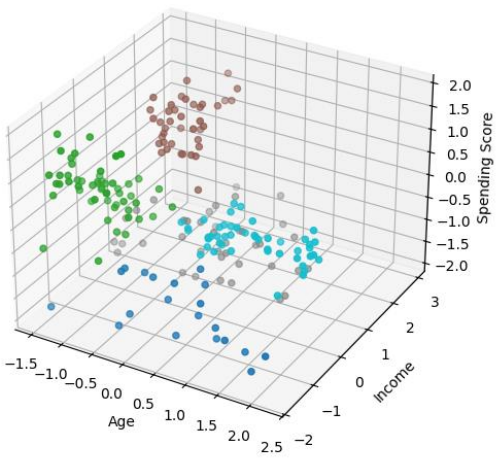
Through profiling of different clusters, behavioural characteristics between clusters can be identified. Clusters sometimes denote high-income high-spending customers and sometimes low-income low-spending customers. This differentiation is crucial for targeted strategies.

Cluster Distribution:

The distribution of customers across clusters shows that some segments are more prominent than others. This information is useful for prioritising business efforts toward larger or more valuable groups.



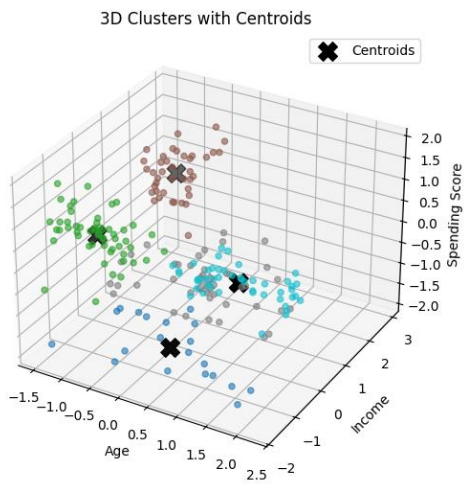
Three-Dimensional Clustering:



The inclusion of age adds depth to the analysis. The 3D visualisation reveals patterns that are not visible in two-dimensional space. It highlights how demographic factors influence spending behaviour.

3D Cluster Centers:

The centroids in three-dimensional space provide a comprehensive summary of each cluster. Their separation confirms that clusters remain distinct even when additional features are included.



Discussion:

The findings show that K-Means clustering which assists with validation and feature scaling is very effective in customer segmentation. Applying both Elbow Method and Silhouette Score for clustering gives assurance that the final solution is robust and not arbitrary.

This study's key strength is the incorporation of multi-dimensional analysis. Segmentation becomes more realistic when age is included in the clustering process of customers' segmentation. It is very important to consider more than one variable in any analysis.

The current clusters can provide useful insights from a practical viewpoint. Businesses can design targeted marketing strategies based upon customer segments. For instance, exclusive service might be extended to premium customers whereas low engagement group might need incentives.

Nonetheless, K-Means has its drawbacks. This method assumes that the clusters are of a similar shape and size; however, in reality, this is not always the case. Moreover, the algorithm is sensitive to initial conditions and therefore different centroid initialisation will yield different results.

Having mentioned those limitations, the structure that this study has assures reliability and interpretable results.

Limitations:

- Assumes clusters are spherical and equally distributed.
- Sensitive to initial centroid placement.
- Limited features may not capture full customer behaviour.
- Overlapping clusters may reduce clarity.
- Results depend on chosen number of clusters.

Conclusion:

This study demonstrates that a structured approach to K-Means clustering can effectively uncover meaningful customer segments.

- The Elbow Method and Silhouette Score jointly identified five clusters as optimal
- Feature scaling ensured accurate distance-based clustering
- Two-dimensional analysis revealed clear behavioural patterns
- Cluster profiling enabled interpretation of customer groups
- Three-dimensional analysis provided deeper insights into demographic influences
- Combining validation techniques with visual analysis improved robustness

Future Work:

- Explore advanced clustering algorithms such as DBSCAN
- Incorporate additional behavioural features
- Apply clustering to real transactional datasets
- Use segmentation results for predictive modelling
- Analyse customer behaviour over time

References:

MacQueen, J. (1967). Some methods for classification of multivariate observations

Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python

Jain, A. K. (2010). Data clustering: 50 years beyond K-Means

Mall Customers Dataset: https://www.kaggle.com/datasets/kandij/mall-customers